

Revisiting Disaggregated LLM Serving for Performance and Energy Implications

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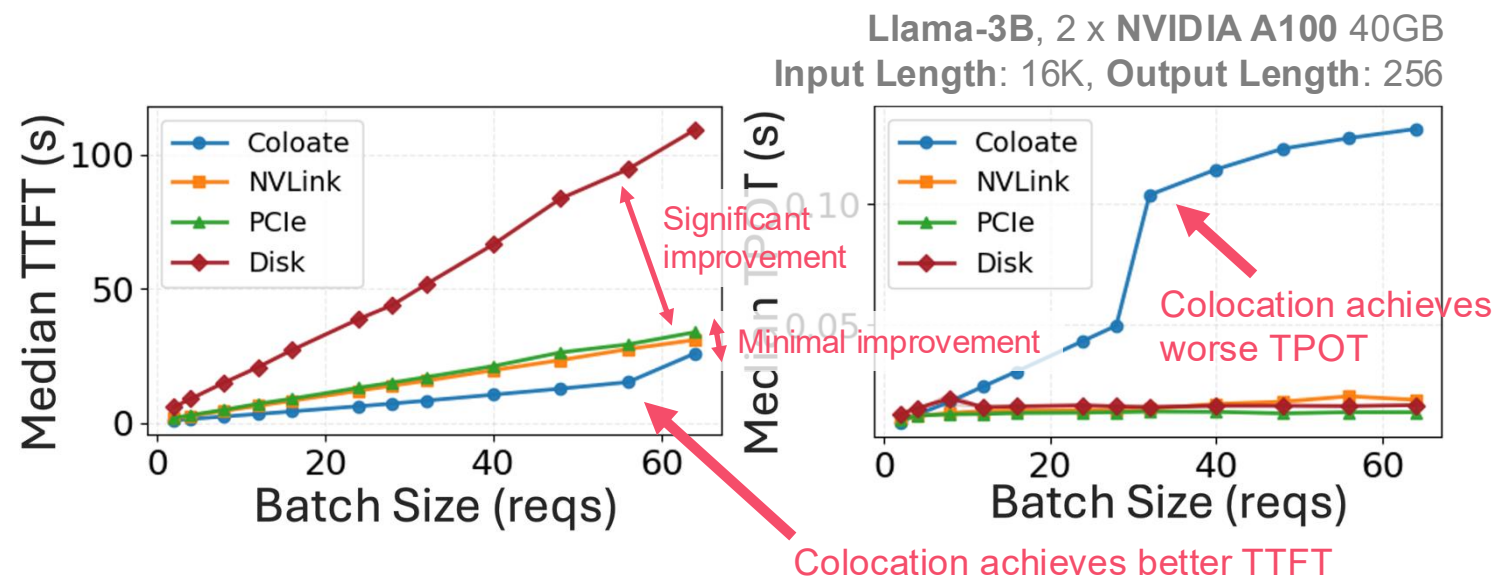
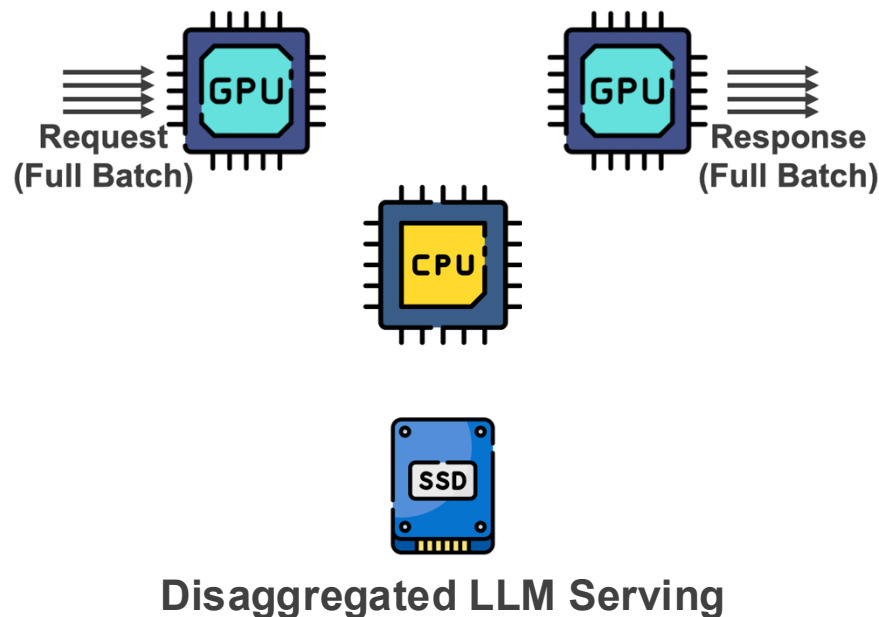
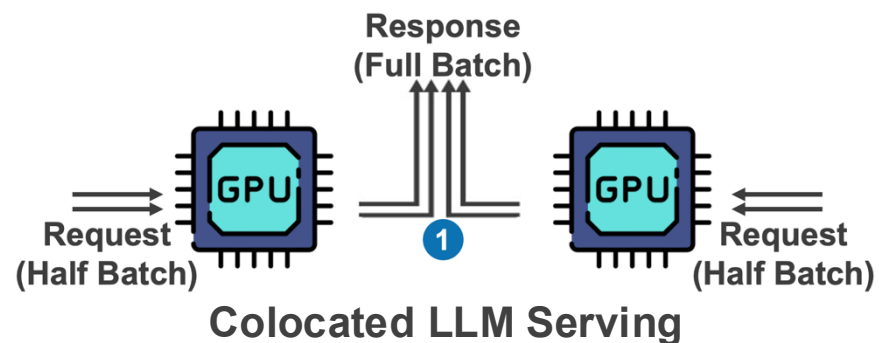
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Benchmark Exp. 1: KV Transfer Paths

Different KV Transfer Paths → Different Serving Performance and Energy Consumption

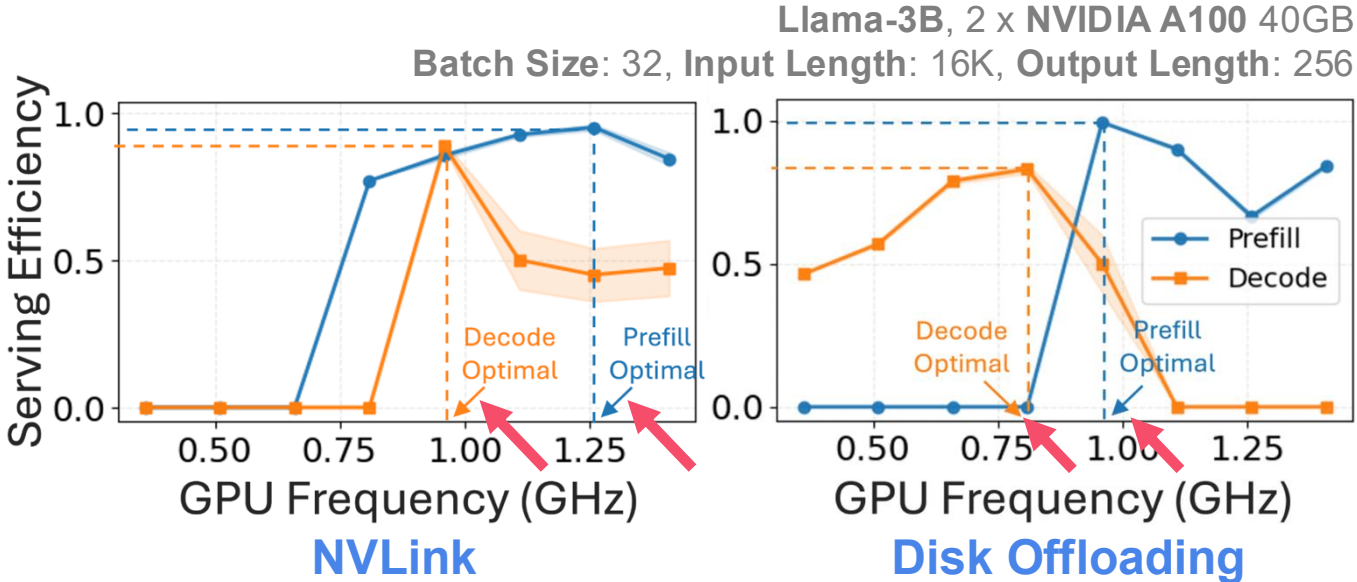
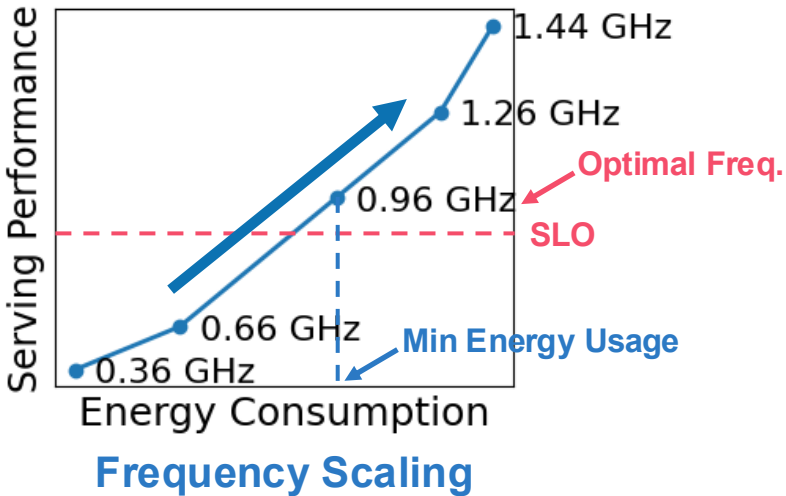


Observation

- Performance benefits of prefill-decode disaggregation are not universal
- KV transmission bandwidth exhibits diminishing returns in performance

Benchmark Exp. 2: Stage-wise Frequency Scaling

Stage-wise Independent Frequency Scaling → Improved Energy Efficiency under SLOs

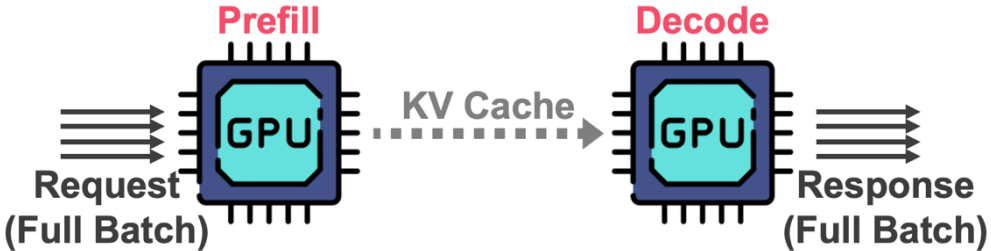


Define: $ServingEfficiency = \begin{cases} \frac{Throughput}{Energy}, & Latency \leq SLO \\ 0, & Latency > SLO \end{cases}$

Observation

- Optimal GPU frequency differs between prefill and decode stages
- Independent frequency scaling further optimizes serving efficiency under SLOs

TTFT SLO → Best Prefill Freq. TPOT SLO → Best Decode Freq.



Disaggregated LLM Serving

Key Takeaway

- Performance gains of prefill–decode disaggregation are not universal
 - Its effectiveness depends on LLM model size, request batch size, and the specific inference stage
- KV transmission bandwidth exhibits diminishing returns in serving performance
- Stage-wise frequency scaling enables improved energy efficiency under SLO constraints